Cisco PE Node Failure SOP - KOR-PE-01 at Koramangala Datacenter

Node: KOR-PE-01 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_9010

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve hardware/environmental faults (ASIC memory parity errors, line card crashes, power supply redundancy loss) on the Cisco PE Node (KOR-PE-01) at the Koramangala Datacenter. The PE Node is a critical component of the network infrastructure, providing high-speed connectivity between the service provider edge and the customer edge. The PE Node's failure can result in service outages, affecting multiple customers. The ASIC memory parity errors are typically caused by faulty or malfunctioning components in the line cards, which can lead to data corruption and system crashes. The line card crashes can be due to overheating, power supply failures, or hardware defects. The power supply redundancy loss can occur due to a faulty power supply unit, leading to a loss of redundancy and potential system failure. The ALARM_9010 alarm code is triggered when the system detects a hardware or environmental fault, indicating a Major severity failure. The following SOP outlines the steps to diagnose and resolve the failure, ensuring minimal downtime and maximum service availability.

Pre-requisite Checks

- Verify that the Koramangala Datacenter is accessible and that the network team has physical access to the KOR-PE-01 PE Node.

- Check the redundant link status to ensure that the network is fully redundant.

- Verify that the power supply units are functioning correctly and that the system is not experiencing any power-related issues.

Potential Root Causes

Cause	Probability
Faulty or malfunctioning components in the line cards	High
Overheating of the line cards	Medium
Power supply failure	Low

Diagnostic Commands & Expected Outputs

```
Command: show tech-support detail, Expected Output: System logs and technical support information, Description: This command collects system logs and technical support information, which can help identify the root cause of the failure. The output will include system configuration, logs, and error messages that can aid in troubleshooting.
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Command: show environment, Expected Output: Environmental parameters such as temperature, humidity, and power supply status, Description: This command displays environmental parameters such as temperature, humidity, and power supply status, which can help identify potential environmental issues that may be contributing to the failure.

Command: show power supply, Expected Output: Power supply status and redundancy information, Description: This command displays power supply status and redundancy information, which can help identify potential power-related issues that may be contributing to the failure.
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Resolution Steps

Step 1: Step: Replace faulty or malfunctioning line cards, Command: no service modular, Description: This command places the line cards in a maintenance state, allowing for their replacement. After replacing the faulty line cards, the network team must verify that the system is functioning correctly and that the ALARM_9010 alarm code is cleared.

Step 2: Step: Verify power supply redundancy, Command: show power supply, Description: This command verifies that the power supply redundancy is functioning correctly and that the system is not experiencing any power-related issues.

Step 3: Step: Verify line card status, Command: show module, Description: This command verifies the status of the line cards, ensuring that they are functioning correctly and that the ALARM_9010 alarm code is cleared.

Post-Check Verification

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Command: show tech-support detail, Expected Output: System logs and technical support information, Description: This command verifies that the system logs and technical support information are updated and reflect the resolution of the failure.

Command: show environment, Expected Output: Environmental parameters such as temperature, humidity, and power supply status, Description: This command verifies that the environmental parameters are within normal operating ranges and that the system is functioning correctly.

Command: show power supply, Expected Output: Power supply status and redundancy information, Description: This command verifies that the power supply status and redundancy information are correct and that the system is functioning correctly.
```

Roll-back Procedures

- Step: Restore original configuration, Command: copy running-config startup-config, Description: This command restores the original configuration, ensuring that the system is returned to its previous state.

- Step: Verify system functionality, Command: show tech-support detail, Description: This command verifies that the system is functioning correctly and that the ALARM_9010 alarm code is cleared.